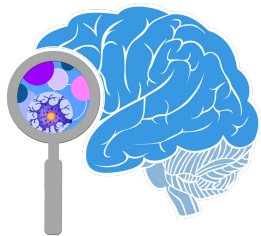


Breathing

Lesson objectives:

- To describe how air moves into and out of the lungs
- To create a scientific model to show how air moves in and out of the lungs



Breathe in... breathe out...

What are the scientific words for what you have just done?

Answer the following questions in your book:



1. What is the function of the respiratory system?

To move air in and out of our lungs

2. What is the name of the small air sacs inside our lungs?

Alveoli

3. Is there any carbon dioxide in the air that we breathe in?

Yes (0.03%)

4. Organise these keywords in the correct order to show the passage of air through our respiratory system

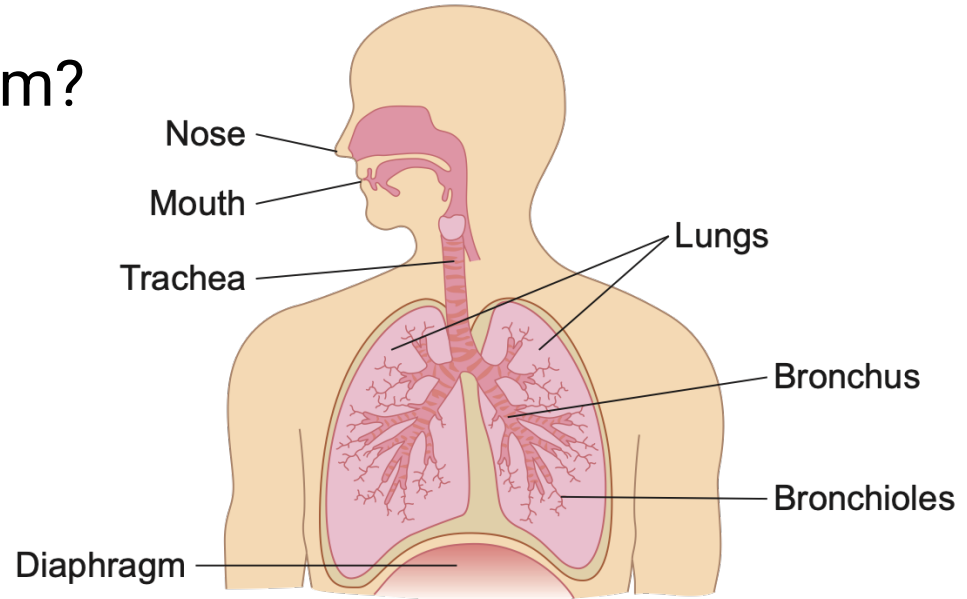
trachea, bronchioles, mouth, nose, alveoli, bronchi

mouth/nose > trachea > bronchi > bronchioles > alveoli

The respiratory system

The lungs cannot make air move in or out. Only muscle can contract/relax to cause movement.

Can you name the muscles related to the respiratory system?



Breathing

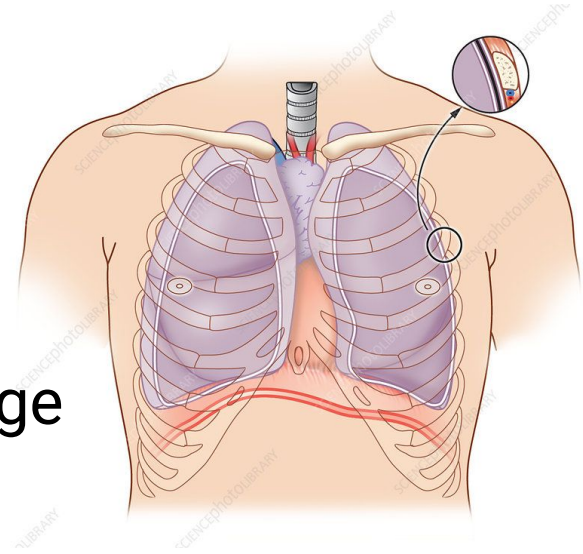
The lungs are surrounded by the **ribcage**.

The ribs have **intercostal muscles** between them.

These contract and relax to move the rib cage

Below the lungs sits a layer of muscle called the **diaphragm**.

The **diaphragm** can contract and relax to change the size of the **chest cavity** (where the lungs are).



How much air can you breathe out?

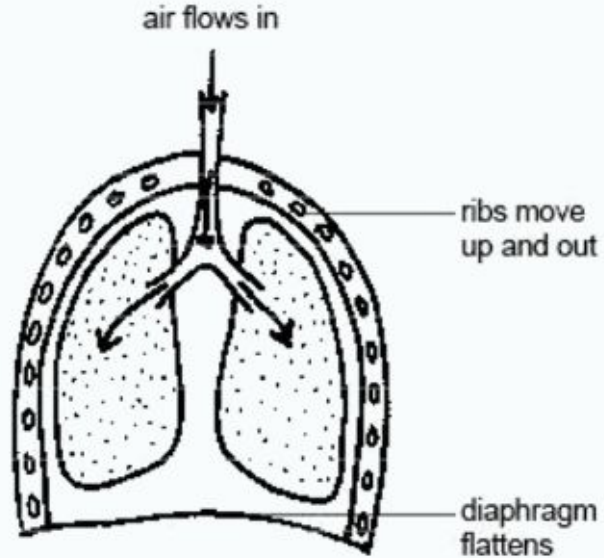


Spirometers are commonly used to measure "**vital capacity**," the **maximum volume of air that you can exhale in one breath**.

Everybody's vital lung capacity is **different** depending on factors such as their **age** and **fitness levels**.

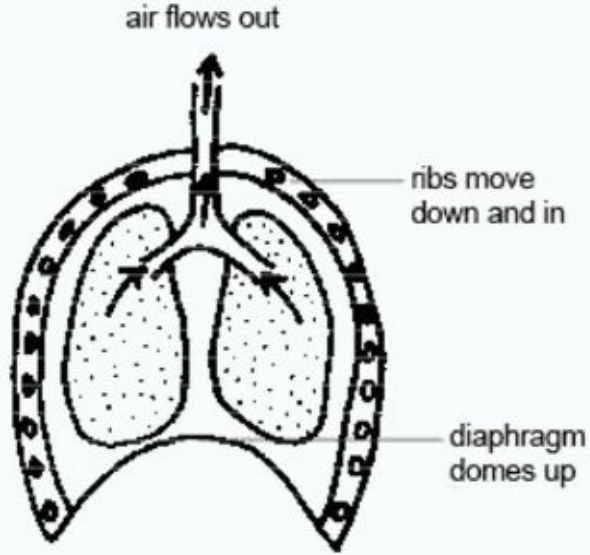


We can also use some very simple apparatus to find out - a plastic bottle, measuring cylinder, bendy tubing and water!



Intercostal muscles contracted

Diaphragm contracted

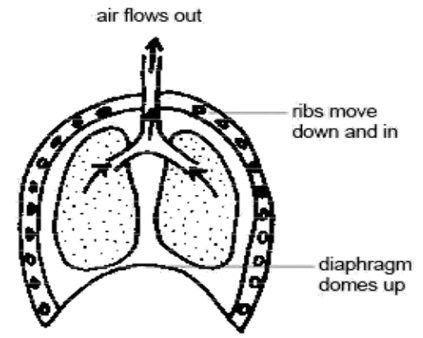
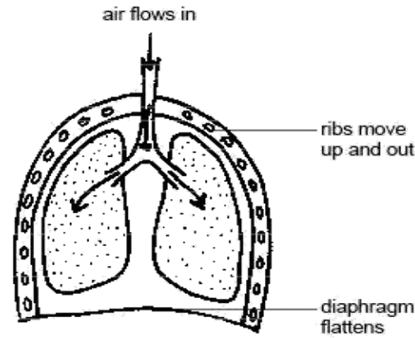


Intercostal muscles relaxed

Diaphragm relaxed

Increase or decrease?

What happens to the size of the chest cavity when:



- a) The intercostal muscles contract
- b) The diaphragm relaxes
- c) The intercostal muscles relax
- d) The diaphragm contracts
- e) The diaphragm is dome-shaped

Using a model to represent breathing movements



1. What are the **benefits** of using a scientific model?

使用科学模型有什么好处？

과학적 모델을 사용하면 어떤 이점이 있나요?

2. Are there any **limitations** to using a scientific model?

使用科学模型有什么限制吗？

과학적 모델을 사용하는 데에는 제한이 있습니까？



Balloon Lung Model



Using a model to represent breathing movements



Work in your groups to create a balloon lung model.

Draw and label the model in your books.

Describe what happens when you move the diaphragm?

What are the **similarities** and **differences** between the bell jar model and our respiratory system?

Discussion Questions (answer in your book in full sentences)

1. Which part of the balloon lung model represents the diaphragm in our body?
What does the diaphragm do when we breathe?
2. Explain the steps you took to make the balloon inflate and deflate. How is this action similar to real breathing?
3. Why did we use 2 balloons for our model? How does this reflect the structure of our actual lungs?
4. Is the balloon lung model a perfect representation of our actual lungs? What are some differences between the model and real lungs that you've learned about through this activity?

Balloon Lung Model

Create a table to show the advantages and limitations of the balloon lung model?

(Can you think of others to add?)

Advantages	Limitations
• Good representation of how air is moved into and out of the lungs • Helps to visualise the mechanism	• The balloons simplify the structure of the lungs as there's no alveoli or bronchioles • Not an accurate size • The rib cage does not move



Describe what happens when we breathe in?

Describe what happens when we breathe out?

TO-DO

Describe how air moves **into** the lungs.

Sentence starters:

- The intercostal muscles...
- The diaphragm...
- This causes the volume of the chest cavity to...
- This means that **air moves**...

To 'describe', your answer should:

- Use **bullet points**.
- Include each step of the process in a **logical order**.
- Use **keywords**.

Key words

Diaphragm, intercostal muscles, contract, increase, decrease

TO-DO

Describe how air moves **into** the lungs.

Model Answer:

- The intercostal muscles of the ribcage **contract** and the ribcage moves up and out.
- The diaphragm **contracts** and **moves downwards to become flat**
- This causes the volume of the chest cavity to **increase**, so the pressure inside the chest cavity **decreases**.
- This means that **air moves down the trachea into the lungs** to fill the extra space

To 'describe', your answer should:

- Use **bullet points**.
- Include each step of the process in a **logical order**.
- Use **keywords**.

Key words

Diaphragm, intercostal muscles, contract, increase, decrease